

METRO TICKET GENERATING SYSTEM IN SERVICENOW

1. INTRODUCTION

1.1 Project Overview

The Metro Ticket Generating System is a ServiceNow-based application developed to digitize and automate the metro ticket booking process. The system enables passengers to book metro tickets online through a Service Catalog, calculate fares automatically, generate QR-code-based digital tickets, and receive instant ticket confirmations.

The solution eliminates manual ticketing challenges, reduces waiting time at metro stations, and provides a convenient paperless travel experience.

1.2 Purpose

The purpose of this project is to:

- Provide a digital platform for metro ticket booking.
- Automate fare calculation and ticket generation.
- Promote cashless transactions.
- Reduce operational effort and paper usage.
- Improve passenger convenience and satisfaction.

2. IDEATION PHASE

2.1 Problem Statement

Metro passengers often experience long waiting queues, delays in ticket purchasing, and inconvenience due to manual ticketing processes. Traditional paper tickets also increase operational costs and reduce efficiency.

The proposed solution aims to provide a digital ticket booking system through ServiceNow that automates ticket generation and improves the overall passenger experience.

2.2 Empathy Map Canvas

Think & Feel

- I want to avoid long queues.

- Ticket booking should be quick and easy.
- I prefer digital payments.
- I do not want to miss my train.

Hear

- Metro stations are crowded during peak hours.
- Digital ticketing is more convenient.
- QR-code tickets provide faster access.

See

- Long queues at ticket counters.
- People using mobile phones for payments.
- QR scanners at metro stations.

Say & Do

- Book metro tickets frequently.
- Use UPI and digital wallets.
- Search for the quickest travel option.

Pain

- Waiting time at ticket counters.
- Manual ticket booking delays.
- Risk of losing paper tickets.

Gain

- Instant ticket booking.
- Digital QR-code tickets.
- Cashless transactions.
- Faster station entry.

2.3 Brainstorming

Identified Ideas

- Online Metro Ticket Booking
- Automated Fare Calculation
- QR Code Ticket Generation
- Digital Payment Integration
- Ticket History Management
- Email Notifications
- Metro Station Mapping
- Reporting & Analytics Dashboard

Selected Idea

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. Passenger accesses Service Portal.
2. Selects source and destination stations.
3. Chooses passenger type and ticket quantity.
4. System calculates fare automatically.
5. Passenger completes payment.
6. QR-code ticket is generated.
7. Passenger receives confirmation.
8. Ticket is validated at metro station.

3.2 Solution Requirement

Functional Requirements

- Ticket Booking
- Fare Calculation
- QR Code Generation
- Digital Ticket Creation
- Email Notification
- Ticket History
- Dashboard & Reports

Non-Functional Requirements

- Security
- Reliability
- Availability
- Scalability
- Performance
- Usability

3.3 Data Flow Diagram

Passenger → Ticket Booking → Station Validation → Fare Calculation → Payment Processing → QR Code Generation → Ticket Storage → Notification Service → Digital Ticket Delivery

3.4 Technology Stack

Component	Technology
Platform	ServiceNow
Frontend	Service Portal, UI Builder
Backend	Business Rules, Flow Designer
Database	ServiceNow Tables
Notification	Email Service
Integration	QR Code API
Security	ACLs, RBAC

4. PROJECT DESIGN

4.1 Problem Solution Fit

Customer Segment

Metro Passengers

Problem

Long queues and manual ticketing process.

Solution

Digital ticket booking through ServiceNow with QR-code ticket generation.

Value Proposition

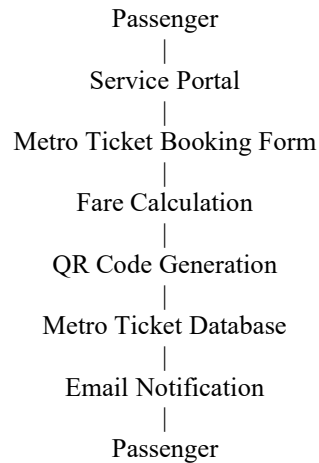
Fast, secure, paperless ticket booking experience.

4.2 Proposed Solution

The Metro Ticket Generating System allows passengers to:

- Book metro tickets online.
- Calculate fares automatically.
- Generate QR-code-based tickets.
- Make digital payments.
- Receive email confirmations.
- View ticket history.

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint 1

- Catalog Form Creation
- Station Mapping
- Ticket Booking Module

Sprint 2

- Fare Calculation Logic
- Payment Integration
- Ticket Generation

Sprint 3

- QR Code Generation
- Email Notifications
- Ticket History

Sprint 4

- Dashboard Development
- Testing
- Deployment

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test Scenario	Result
Ticket Booking	Pass
Fare Calculation	Pass
QR Generation	Pass
Email Notification	Pass
Dashboard Reports	Pass

Test Summary

- Total Test Cases: 60
- Passed: 59
- Failed: 1
- Success Rate: 98.33%

7. RESULTS

7.1 Output

The screenshot displays the ServiceNow Studio interface for the 'Metro Ticketing System' app. The left sidebar shows the app's structure with sections like Security, Role, and Other. The main area shows the app details, including the scope 'x_1971522_metro_0' and the date created '2026-06-22 15:27:15'. Below this, there's a table listing files with columns for Name, Type, Updated, Updated by, and Created by. The table is currently empty, showing only headers and a 'No data yet' message. At the bottom, a dialog box titled 'Now, let's get more info about your new table' is open, prompting the user to define the properties of a new table. The dialog includes fields for 'Table label', 'Table name prefix', and 'Table name', along with checkboxes for 'Make extensible' and 'Auto number'.

ServiceNow Studio | Metro Ticketing System | Automated digital ticket generation and validation system

App type: Custom | Scope: x_1971522_metro_0 | Date created: 2026-06-22 15:27:15 | sys_id: 0874498783ad07148cad9420cead3d8

Files | Filter list | Updated (Newest)

Name	Type	Updated	Updated by	Created by
No data yet Data is the starting point for any app. Start with a table or upload a spreadsheet or PDF.				
No user interface yet Start on the front-end by adding an interface, like a workspace or portal.				
No automation yet Automation streamlines processes and workflows within your organization.				
No security yet Security streamlines processes and workflows within your organization.				

Now, let's get more info about your new table

Define the properties of your new table.

Table label *

Table name prefix * Table name *

☐ Make extensible ☐ Auto number

Cancel Continue

ServiceNow Studio

Metro Ticketing System

New Table

Apps

Filter list

Metro Ticketing System

MKCE Connect

Source

Define

Summary

CREATE TABLE

Your table is ready

You can continue to edit this table or get back to building the rest of your app.

Source

Table name

Fields

Blank

Metro Station's Details

0

Edit table

Done

servicenow

All

Favorites

History

Workspaces

Admin

Tables

Search

Table Metro Station's Details

Delete

Update

Delete All Records

* Label

Metro Station's Details

* Name

x_1971522_metro_0_metro_station_s_details

Extends table

Application

Metro Ticketing System

Create module

Create mobile module

Add module to menu

-- Create new --

New menu name

Remote Table

Columns

Controls

Application Access

Table Columns

for text

Search

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display	Updated	Updated by	Application	Created
Updated by	String	(empty)	40		false	2026-06-22 08:33:07	admin	(empty)	2026-06-22 08:33:07
Updates	Integer	(empty)	40		false	2026-06-22 08:33:07	admin	(empty)	2026-06-22 08:33:07
Updated	Date/Time	(empty)	40		false	2026-06-22 08:33:07	admin	(empty)	2026-06-22 08:33:07
Created by	String	(empty)	40		false	2026-06-22 08:33:07	admin	(empty)	2026-06-22 08:33:07
Sys ID	String	(empty)	32		false	2026-06-22 08:33:07	admin	(empty)	2026-06-22 08:33:07

ServiceNow Studio

Metro Ticketing System

Metro Station's Details - Del

Table Metro Ticketing System

App details

Filter list

Metro Station's Details

Table Station's Details

Table fields

Add new field

Column label *	Column name * IF	Type *	Reference	Max length
Updated	sys_updated_on	Date/Time		
Updated by	sys_updated_by	String		40
Updates	sys_mod_count	Integer		
Created	sys_created_on	Date/Time		
Created by	sys_created_by	String		40
Station Name	station_name	String		255

Station Name

station_name

String

Field details

Label

Station Name

Type *

String

Read only

Mandatory

Active

Max. length

255

Default value

Use dynamic default

More properties available

Showing 1-5 of 5

1

Scope: Scope is controlled & indicated by the builder.

ServiceNow Studio

Metro Ticketing System | Metro Station's Details - De... | Metro Station's Details - De... | Metro Station's Details - De...

App details | Policies and rules | Metro Ticketin... | Preview | Save

Filter list

- Data
 - Form Section (1)
 - (empty)
 - Table (1)
 - Metro Station's Details
 - Table Column (1)
- Security
 - Access Section (8)
 - Role (2)
- User Interface
 - List (1)
- Other
 - Access Roles (8)
 - Embedded Help Role Pri... (2)
 - Field Label (1)
 - Table Subscription Conf... (1)

Add form elements

Fields (4) | Components (2) | More

+ Add a field in the table

- Created Date/Time
- Created by String
- Updated by String
- Updates Integer

Default view | Metro Station's Details | One-column

Metro Station's Details

Station Name

Station Code

Zone

Is Active

Updated 2008-12-06 18:47:00

Section

Section label: Metro Station's Details

Section layout: One column | Two columns

Merge with section above

Learn more about sections

You can merge multiple sections together to create a single section with different layouts and one section label.

When you merge a section, it combines with the section directly above it. This also applies if you add a new section above a merged section.

Scope: Scope is controlled & indicated by the builder.

Catalog Item

Book A Metro Ticket Item

Build and modify items faster with the improved Catalog Builder.

Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

Name: Book A Metro Ticket Item

Application: Metro Ticketing System

Catalogs: Service Catalog

Category: Services

State: -- None --

Checked out: -- None --

Owner: System Administrator

Active: ☒

Fulfillment automation level: Unspecified

Item Details | Process Engine | Picture | Pricing | Portal Settings

Short description

Description

B I U ↶ ↷ Verdana 8pt

Workflow Studio

Metro Project | Metro Ticketing System

Metro Project | Inactive

View: Test | Activate

TRIGGER

- Service Catalog

ACTIONS | Select multiple

- Get Catalog Variables from Book A Metro Ticket Item
- Create Metro Station's Details Record

Add an Action, Flow Logic, or Subflow

ERROR HANDLER

If an error occurs in your flow, the actions you add here will run.

Data | Collapse All

- Flow Variables
- Trigger - Service Catalog
 - Requested Item Record (Record)
 - Run Start Time UTC (Date/Time)
 - Run Start Date/Time (Date/Time)
 - Table Name (Table Name)
- 1 - Get Catalog Variables
- 2 - Create Record
 - Metro Station's Details Record (Record)
 - Metro Station's Details Table (Table)
 - Action Status (Object)

Pre-configured ServiceNow actions are not editable

Book A Metro Ticket Item

Press Alt+0 for help

Add relevant tags to the Meta field using comma-separated list of tags. These tags will be used while searching the item. Not applicable if AI Search is configured.

Meta

Copy

Try It

Update

Edit in Catalog Builder

Delete

Related Links

[Item Diagnostic](#)

[Run Point Scan](#)

[\[SN Utils\] Versions \(4\)](#)

Variables

Variable Sets

Catalog UI Policies (1)

Catalog Client Scripts (1)

Available For

Not Available For

Categories (1)

Catalogs (1)

Catalog Data Lookup Definitions

Related Articles

Related Catalog Items

Assigned Topics

Order

Search

Actions on selected rows...

New

Catalog item = Book A Metro Ticket Item

☐	Short description	Variable set	Conditions	Reverse if false	On load	Inherit	Updated	Order
☐	Book A Metro Ticket Item	(empty)		true	true	false	2026-06-27 11:11:13	100

1 to 1 of 1

Copy

Try It

Update

Edit in Catalog Builder

Delete

Related Links

[Item Diagnostic](#)

[Run Point Scan](#)

[\[SN Utils\] Versions \(4\)](#)

Variables

Variable Sets

Catalog UI Policies (1)

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Catalog Data Lookup Definitions

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☐	Book A Metro Ticket Item	(empty)		true	true	false	2026-06-27 11:11:13	100

1 to 1 of 1

Copy

Try It

Update

Edit in Catalog Builder

Delete

Related Links

[Item Diagnostic](#)

[Show VA render type](#)

[Run Point Scan](#)

[\[SN Utils\] Versions \(8\)](#)

Variables

Variable Sets

Catalog UI Policies (1)

Catalog Client Scripts (3)

Available For

Not Available For

Categories (1)

Catalogs (1)

Catalog Data Lookup Definitions

Related Articles

Related Catalog Items

Assigned Topics

for text

Search

Actions on selected rows...

New

Catalog item = Book A Metro Ticket Item

☐	Name	Active	Table	View	Type	Updated
☐	Validation	true			onChange	2026-06-28 00:36:15
☐	QR Generation	true			onSubmit	2026-06-28 00:34:13
☐	FareCalculation	true			onChange	2026-06-27 10:58:15

1 to 3 of 3

Copy

Try It

Update

Edit in Catalog Builder

Delete

Build and modify items faster with the improved Catalog Builder.

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Enter a Name and Short description to display for the item.

Enter a Price, approvals, variables, and other information as needed.

Name

Book A Metro Ticket Item

Application

Metro Ticketing System

Catalogs

> Service Catalog

Active

☒

Category

Services

Fulfillment automation level

Unspecified

State

-- None --

Checked out

-- None --

Owner

System Administrator

Item Details

Process Engine

Picture

Pricing

Portal Settings

Short description

Description

Catalog Item
New record

Build and modify items faster with the improved Catalog Builder.

Catalog Items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

NameBook A Metro Ticket

ApplicationMetro Ticketing System

CatalogsService Catalog

Active

CategoryServices

Fulfillment automation levelUnspecified

State-- None --

Checked out-- None --

OwnerSystem Administrator

Item Details

Process Engine

Picture

Pricing

Portal Settings

Short descriptionBook A Metro Ticket

Description

Catalog Item
Book A Metro Ticket

CopyTry ItUpdateEdit in Catalog BuilderDelete

Related Links

Item Diagnostic

Run Point Scan

SN Utils Versions (7)

Variables (4)

Variable Sets

Catalog UI Policies

Catalog Client Scripts

Available For

Not Available For

Categories (1)

Catalogs (1)

Catalog Data Lookup Definitions

Related Articles

Related Catalog Items

Assigned Topics

Order

Search

Actions on selected rows...

New

Catalog Item = Book A Metro Ticket

Type	Question	Order
Single Line Text	Enter Smart card number	100
Multiple Choice	What do you want today?	200
Multiple Choice	Type of Journey	300
Reference	Starting from	400

Order Confirmation

Request forAbel Tuter

Delivery Information (Optional)

Special instructions (Optional)

CancelSubmitting...

Book A Metro Ticket

Book A Metro Ticket

*Enter Smart card number695898

What do you want today?

Type of Journey

Starting from

Choose a file or drag it here, Copy and paste clipboard files here.

HomeRequest Summary - REQ0010001

Search Catalog

Submitted: 2026-06-28 00:51:43

Request Number: REQ0010001

Requested for: Abel Tuter

Estimated Delivery: 2026-06-30

Item	Delivery Date	Stage	Price (each)	Quantity	Total
Book A Metro Ticket	2026-06-30	Request Approval	---	1	---

Total: \$0.00

Home
>
Service Catalog
>
Office
>
Services
>
Book A Metro Ticket

Search Catalog

Book A Metro Ticket

Book A Metro Ticket

Indicates required

Enter Smart card number

Where do you want to go?

☒ Chennai
☐ Coimbatore
☐ Salem

Type of Journey

☒ None
☐ Return[add \$200.00]
☐ Single[add \$200.00]
☐ Up and Down[add \$400.00]

Starting from

Quantity: 1

Price: \$100.00

Delivery Time: 2 Days

Add to Cart

Save as Draft

Order Now

Required Information

Enter Smart card number

Home
>
Request Summary - REQ0010003

Search Catalog

Submitted :2026-06-28 02:06:45
Request Number : **REQ0010003**
Estimated Delivery : 2026-06-30

Item	Delivery Date	Stage	Price (each)	Quantity	Total
Book A Metro Ticket	2026-06-30	Assess or Scope Task	\$100.00	1	\$100.00
					Total: \$100.00

Home
>
My Request - RITM0010003

BOOK A METRO TICKET

Item
Book A Metro Ticket

Requested for
System Administrator

Quantity
1

Price
\$100.00

Stage
Assess or Scope Task

Activity
Attachments
Additional Details

Type your message here...
Post

System Administrator
RITM0010003 Created

1m ago

Start

8. ADVANTAGES & DISADVANTAGES

Advantages

- Paperless ticketing system.
- Faster ticket booking process.
- Improved passenger convenience.
- Secure digital payments.
- Reduced operational effort.
- Better reporting and analytics.

Disadvantages

- Requires internet connectivity.
- Dependent on ServiceNow availability.

- QR scanner dependency at stations.

9. CONCLUSION

The Metro Ticket Generating System successfully digitizes the metro ticket booking process through ServiceNow. The system automates fare calculation, ticket generation, and passenger notifications, thereby improving commuter experience and operational efficiency while promoting paperless transportation services.

10. FUTURE SCOPE

- Mobile App Integration
- WhatsApp Ticket Booking
- Smart Card Recharge
- Real-Time Train Tracking
- AI-Based Passenger Analytics
- Multi-Language Support
- Dynamic Fare Calculation

11. APPENDIX

GitHub Repository Link

<https://github.com/HemaSuthakar/Metro-Ticket-Generating-System-in-ServiceNow.git>

Project Demo Link

https://github.com/HemaSuthakar/Metro-Ticket-Generating-System-in-ServiceNow/blob/main/7.Project%20Demonstration/1000073224_1918x1002.mp4